scientific reports

OPEN

Soybean selection in Kenya enhanced by multi-trait and genotype-by-environment interaction modeling

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The growing global demand for soybean protein is driving the expansion of cultivation into new agricultural frontiers. Kenya has been progressing in the development of soybean genotypes to identify those best adapted to its diverse agroecological conditions. However, the selection of genotypes with superior agronomic traits and high stability remains limited. This study quantified the magnitude of genotype-by-environment (G×E) interaction and selected genotypes with the best performance and stability based on a multi-trait approach. A total of 65 genotypes were evaluated across 19 environments from 2019 to 2023/24 growing seasons using a randomized complete block design. Stability was assessed using the Weighted Average of Absolute Scores index, considering plant height (PH), number of days to maturity (NDM), and grain yield (GY). Multi-trait selection was performed using the Desired Gain Index, applying a 20% selection intensity, followed by the genetic gain estimation. Using the likelihood ratio test, we identified significant effects of genotype, environment, and G×E interaction. The overall mean values observed in the experiments were 62.30 cm for PH, 120 days for NDM, and 1783.70 kg ha^{-1} for GY. In the multi-trait analysis, we selected the ideotypes G42, G26, G46, G35, G53, G41, G12, G54, G39, G52, G02, and G37. This selection resulted in a genetic gain of 6.5% for PH, 1.3% for NDM, and 15.6% for GY. These genotypes exhibited high genetic potential and adaptation to Kenyan conditions.

Keywords *Glycine max* (L.)., Adaptation, Crossover interaction, Genetic gain, Selection index

Soybean (*Glycine max* (L.) Merr.) is one of the most important global crops, widely cultivated for vegetable oil, protein production, and animal feed[1–3]. In the recent years, soybean expansion has reached tropical and subtropical regions of Africa[4], driven by initiatives such as the Pan-African Soybean Trials Network, which aims to evaluate and promote varieties adapted to local conditions[5]. In Kenya, soybean has emerged as an important alternative for agricultural diversification and a food security crop, highlighting the need to develop genotypes adapted to local edapho-climatic conditions. Conducting multi-environment trials (METs) is essential to identify genotypes with high performance and stability, as these can capitalize on genotype-by-environment (G×E) interaction across different scenarios[6,7].

One of the main challenges breeders face in soybean improvement is the G×E interaction, especially when selecting and recommending cultivars. This interaction can significantly alter the phenotypic expression of soybean traits, such as yield, disease resistance, and tolerance to abiotic stress. When G×E interaction is at crossover, genotype rankings vary considerably across environments, complicating the precise identification of superior materials[8,9]. Therefore, quantifying and comprehensively understanding G×E interactions facilitates the development of more efficient strategies, either through defining mega-environments for targeted selection

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or broad selection aiming at cultivars with consistently good performance across diverse environmental conditions[8].

An effective way to investigate G×E interaction is through reaction norm analysis, which models the phenotypic response of genotypes across an environmental gradient and allows identification of both plasticity and stability patterns[10]. For proper modeling of this interaction, approaches such as mixed linear models (MLM) are widely used to efficiently address these imbalances, enabling improved genetic evaluations and accurate predictions. MLMs effectively handle unbalanced data, optimize genetic value predictions, and capture different types of G×E expression, such as parallel responses (no interaction) or changes in scale and rank across environments[11,12]. They are also effective for multi-trait analyses, supporting more comprehensive genotype evaluations, especially when analyzing large datasets[11].

Traditionally, soybean breeding programs have prioritized single-trait selection strategies, which often overlook the complexity of improving multiple traits simultaneously. This simplification can limit overall genetic gains, especially when traits differ in scale, heritability, and economic relevance[13]. To overcome these limitations, several multi-trait selection approaches have been developed, including the classical Smith-Hazel index[14], the Pesek-Baker index[15], and the multi-trait genotype-ideotype distance index (MGIDI)[16]. These methods allow for the integration of phenotypic data across environments and facilitate the identification of superior genotypes based on multiple selection criteria. However, reducing the subjectivity involved in assigning economic weights remains a key challenge. In this context, approaches such as the Desired Gain Index are essential, as they enable breeders to explicitly define the expected genetic gains for each trait, aligning selection more closely with breeding objectives.

With the support of Pan-African Soybean Trials Network, several African countries have either begun cultivating soybean or significantly expanded their evaluation areas to identify superior genotypes adapted to local conditions[5,17]. Given Kenya's recent expansion and limited history of soybean cultivation, it is necessary to conduct targeted studies to identify high-yielding and stable genotypes. Therefore, this study aimed to: (i) quantify the magnitude of the genotype-by-environment (G×E) interaction for soybean genotypes cultivated under Kenya's edaphoclimatic conditions; and (ii) perform simultaneous selection for multiple traits using data from multiple environments and cropping seasons.

Methods

Experimental site characterization and description

Figure 1A sets the geographical context by mapping Africa's biomes and pinpointing Kenya in the eastern part of the continent. The experimental sites in this study span distinct ecological zones with diverse environmental conditions. Busia and Siaya, in western Kenya, lie within the Tropical and Subtropical Moist Broadleaf Forest biome, which is characterized by high humidity, fertile soils, and evenly distributed rainfall favorable for agricultural practices. Nakuru and Njoro, situated centrally, fall within the Tropical and Subtropical Grasslands and Savannas biome, presenting fertile volcanic soils and seasonal climatic conditions conducive to agricultural experiments. Murang'a, Kirinyaga, and Kiambu, located in high-altitude regions, belong to the Montane Grasslands and Shrublands biome, which is characterized by moderate temperatures and soils rich in organic matter. Kibwezi and Taita Taveta, in the southeast, are positioned in transitional zones between humid biomes and savannas, exhibiting climatic variability that affects agricultural performance. Environmental characterization is supplemented by soil data (Fig. 1B). Figure 1C shows the delimitation of Kenya and its biomes, while Fig. 1D presents the altitude map. All maps were generated in R software (R Core Team, 2025; version 4.5.1)[18], using the packages *sf*[19], *terra*[20], *ggplot2*[21], and *rnaturalearth*[22].

Plant materials

The soybean genotype trials used in this study are part of the Pan-African Trials Network, which aims to accelerate the introduction of test varieties across Africa, the USA, Australia, and Latin America, thereby providing farmers with a broader seed selection. These genotypes, managed under the coordination of SIL[17], consisted of cultivars and experimental lines developed by both public and private breeding programs from Africa, Brazil, and the United States. Field evaluations were conducted in Kenya during the cropping seasons from 2019 to 2023/2024, through international collaborations focused on identifying genotypes adapted to the diverse target population of environments (TPE) across the African continent. A total of 65 unique soybean genotypes (G001 to G065) were tested across nine multi-environment trial sites located mainly in western Kenya (Supplementary Table 1). It is important to note that the number 65 refers to the cumulative set of genotypes evaluated throughout the entire study period; each individual trial typically included a subset of approximately 30 genotypes, depending on the year and location. Some genotypes were discontinued over time due to poor adaptation to specific environments, making them unsuitable to remain in the Pan-African network. The trials were designed to capture agroecological variability representative of key soybean-producing regions in the country.

Experimental design and data collection

Each environment was defined by a combination of location and year, resulting in 19 trials (Supplementary Figure 1). The trials were conducted using a randomized complete block design (RCBD) with three replications and the experimental plots comprised of four (4) rows, each 5 m long, spaced at 0.5 m apart, with 25 plants per meter, resulting in a plot area of 10 m^2. Data were collected in the two central rows, corresponding to a net plot area of 5 m^2, for the agronomic traits; plant height at maturity (PH, cm), measured from the base of the stem to the apex of the main stem; number of days to maturity (NDM, days), recorded as the number of days from sowing to physiological maturity; and grain yield (GY, kg ha^{-1}), calculated from the total grain mass harvested

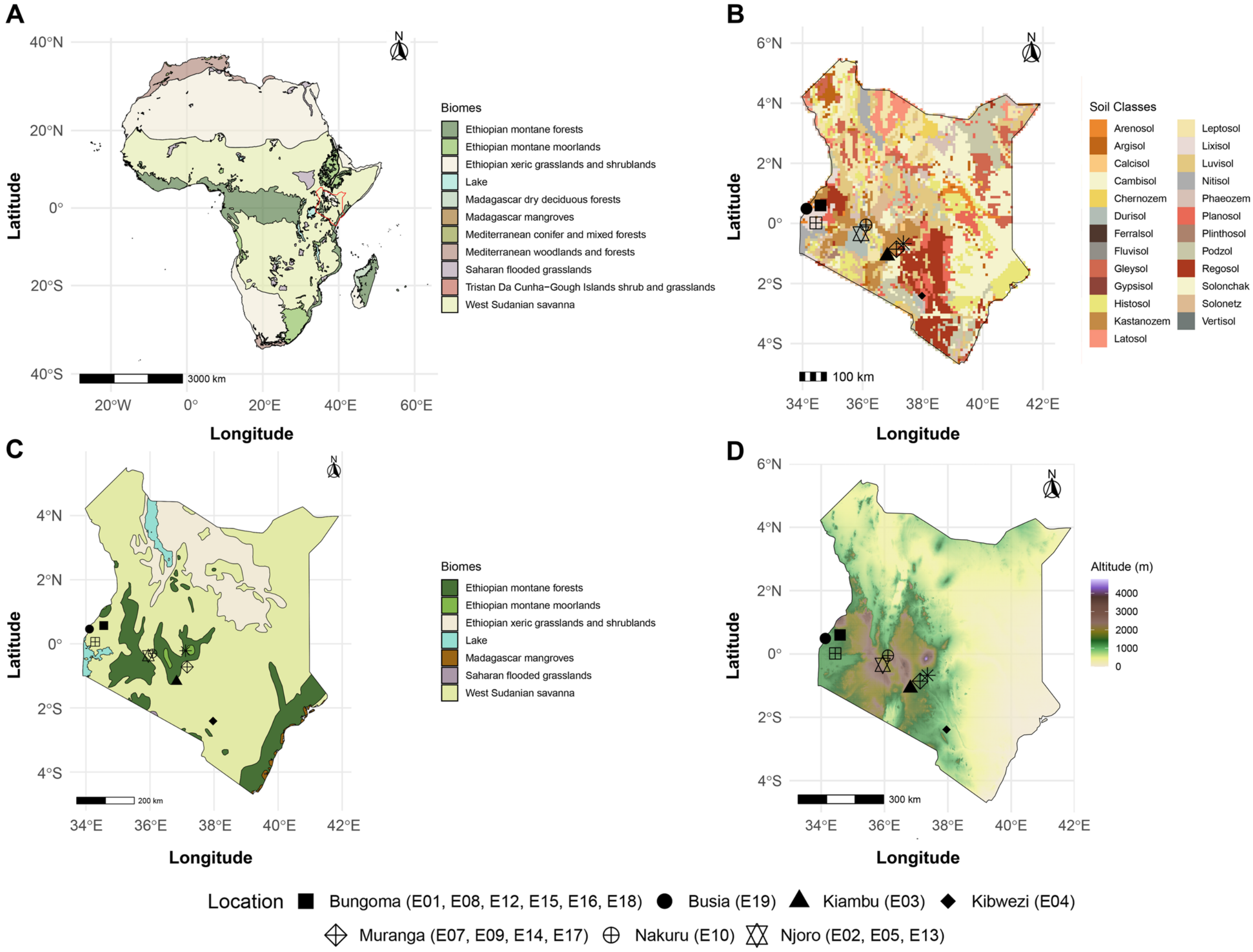

Fig. 1. (**A**) Shows the map of Africa with biome distributions across its entire continent, highlighting Kenya in East Africa. (**B**) Presents the soil classifications in Kenya, emphasizing soil variability in the studied regions. (**C**) Illustrates the diversity of biomes within Kenya. (**D**) Depicts Kenya's altitudinal ranges, distinguishing lowland areas, transition zones, and mountainous regions, which directly influence climatic conditions and agricultural suitability. The geometric shapes represent the locations evaluated in Kenya, and the information in parentheses indicates the number of trials conducted at each location. All maps were generated in *R* Software.

from the net plot area, adjusted to 13% moisture content. Pest and disease management followed standard agronomic recommendations for soybean production.

Statistical analysis

Estimates of variance components for each evaluated trait were obtained by the Restricted Maximum Likelihood (REML) method[23]. Prediction of genotypic values was carried out using Best Linear Unbiased Prediction (BLUP)[24], accounting for the variance-covariance structure of the trials.

Single-environment trial analysis

The analysis of single-environment trials was performed using the following linear mixed model:

$$\mathrm{y} = \mu \mathrm{1} + \mathrm{Xb} + \mathrm{Zg} + \boldsymbol{\epsilon} \tag{1}$$

where y is the vector of phenotypic observations $(n \times 1)$; μ is the intercept; 1 is a vector of ones with dimension $(n \times 1)$; b is the vector of fixed replication effects, with incidence matrix X of dimension $(n \times p)$, where p is the number of levels of the fixed effect; g is the vector of random genotype effects, assumed as $\mathrm{g} \sim N(0, \mathrm{I}\sigma_g^2)$, where σ_g^2 is the genotypic variance and I is an identity matrix of appropriate dimension; the incidence matrix Z has dimension $(n \times v)$, with v being the number of genotypes. The residual errors are represented by $\boldsymbol{\epsilon} \sim N(0, \mathrm{I}\sigma_\epsilon^2)$, where σ_ϵ^2 is the residual variance, with dimension $(n \times 1)$. This model was fitted independently for each environment and trait to estimate the genetic effects within specific trial conditions.

We applied the likelihood ratio test (LRT)[25] to assess the significance of genotype effects according to Eq. (2):

$$LRT = -2(Log\mathrm{L} - Log\mathrm{L}_f) \tag{2}$$

where L represents the log-likelihood of the reduced model, and L_f corresponds to the log-likelihood of the full model. The LRT value was compared against the chi-square distribution.

Multi-environment trial analysis

Analysis of multi-environment trials (MET) was performed using the following model (Eq. (3)):

$$\mathrm{y} = \mu 1_n + \mathrm{X}_1\mathrm{a} + \mathrm{X}_2\mathrm{b} + \mathrm{Zg} + \boldsymbol{\epsilon} \tag{3}$$

where y is the vector of phenotypic observations $(n \times 1)$; μ is the intercept; 1_n is a vector of ones with dimension $(n \times 1)$; a is the vector of fixed environmental effects $(a \times 1)$, with incidence matrix X_1 of dimension $(n \times a)$; b is the vector of fixed replication effects nested within environments $(pa \times 1)$, with incidence matrix X_2 of dimension $(n \times p)$; g is the vector of random genotypic effects within environments, assumed as $\mathrm{g} \sim MVN(0, \Sigma_g \otimes \mathrm{I}_v)$, where Σ_g is the genetic covariance matrix across environments with dimension $(g \times g)$, and I_v is an identity matrix of order v, with v representing the number of genotypes. The corresponding incidence matrix Z has dimension $(n \times v)$; $\boldsymbol{\epsilon}$ is the vector of residual errors, assumed as $\boldsymbol{\epsilon} \sim MVN(0, \Sigma_\epsilon \otimes \mathrm{I}_n)$, where Σ_ϵ is the residual covariance matrix with dimension $(n \times n)$, and I_n is the identity matrix of dimension $(n \times n)$. *n* is the number of observations, *p* the number of replications, *v* the number of genotypes, and *a* the number of environments. This model was fitted separately for each evaluated phenotypic trait (PH, NDM, and GY), enabling the estimation of genotype effects and G×E interaction across the conducted trials.

Broad-sense heritability (H_a^2) was estimated using Eq. (4).

$$H_a^2 = \frac{\hat{\sigma}_g^2}{\hat{\sigma}_g^2 + \hat{\sigma}_{ge}^2 + \hat{\sigma}_\epsilon^2} \tag{4}$$

where $\hat{\sigma}_g^2$ represents the genetic variance, $\hat{\sigma}_{ge}^2$ corresponds to the G×E interaction variance, and $\hat{\sigma}_\epsilon^2$ denotes the residual variance, all estimated from Eq. (3).

The coefficient of determination for G×E interaction effects (r_{ge}^2) was calculated using:

$$r_{ge}^2 = \frac{\hat{\sigma}_{ge}^2}{\hat{\sigma}_g^2 + \hat{\sigma}_{ge}^2 + \hat{\sigma}_\epsilon^2} \tag{5}$$

The G×E correlation (r_{ge}) was estimated using:

$$r_{ge} = \frac{\hat{\sigma}_g^2}{\hat{\sigma}_g^2 + \hat{\sigma}_{ge}^2} \tag{6}$$

The genotypic coefficient of variation (CV_g) and the residual coefficient of variation (CV_r) were estimated using:

$$CV_g = \frac{\sqrt{\hat{\sigma}_g^2}}{\hat{\mu}} \times 100, \quad CV_r = \frac{\sqrt{\hat{\sigma}_\epsilon^2}}{\hat{\mu}} \times 100 \tag{7}$$

where $\hat{\mu}$ represents the overall mean of the trials.

The CV ratio (coefficient of variation ratio) was used to assess the magnitude of genetic variation relative to residual variation. This ratio is defined as:

$$CV_{ratio} = \frac{CV_g}{CV_r} \tag{8}$$

The genetic correlation ($\hat{r}_g$) between traits was estimated using the BLUPs of plant height (PH), number of days to maturity (NDM), and grain yield (GY). The correlation was calculated as:

$$\hat{r}_g = \frac{\hat{\sigma}_{gXY}}{\sqrt{\hat{\sigma}_{gX}^2 \hat{\sigma}_{gY}^2}} \tag{9}$$

where $\hat{r}_g$ = estimated genetic correlation between traits X and Y; $\hat{\sigma}_{gXY}$ = genetic covariance between X and Y; $\hat{\sigma}_{gX}^2$ = genetic variance of trait X; $\hat{\sigma}_{gY}^2$ = genetic variance of trait Y.

We estimated the stability of each genotype using the weighted average of absolute scores. These scores were derived from the singular value decomposition of the matrix containing the BLUPs for the G×E interaction effects[26]. The WAASB_i was computed as:

$$\mathrm{WAASB}_i = \frac{\sum_{k=1}^{p} |\mathrm{IPCA}_{ik} \times \mathrm{EP}_k|}{\sum_{k=1}^{p} \mathrm{EP}_k} \tag{10}$$

where WAASB_i represents the weighted average for the i^{th} genotype. In this equation, IPCA_{ik} corresponds to the score of the i^{th} genotype on the k^{th} interaction principal component axis (IPCA), while EP_k denotes the proportion of variance explained by the kth IPCA. Lower WAASB_i values indicate higher stability.

Olivoto et al.[26] proposed the WAASBY index, which enables simultaneous selection by weighting mean performance (Y) and stability (WAASB_i) based on Eq. (11):

$$\text{WAASBY}_i = \frac{(rY_i \times \theta_Y) + (rW_i \times \theta_S)}{\theta_Y + \theta_S} \tag{11}$$

where $WAASBY_i$ is the simultaneous selection index for the ith variety, balancing mean performance and stability. The parameters θ_Y and θ_S represent the weights assigned to the response variable and WAASB_i, respectively. The variables rY_i and rW_i correspond to the rescaled values of the response variable and WAASB_i, respectively. The analysis was conducted using `R` Software[18], employing the `Metan` package[26].

Desired gain selection index

The Desired Gain Index was proposed by Pesek and Baker[27] to achieve predetermined gains for a set of traits while optimizing the overall selection response. Let n be the total number of traits included in the index, and j be the number of traits for which a targeted gain is desired. The phenotypic variance matrix of the traits in the index is represented as $\text{var}(X) = P_{n\times n}$, while the covariance between X and the genetic values of the target traits is given by $\text{cov}(X, A) = G_{n\times j}$. Thus, the fundamental relationship of the index can be expressed as:

$$\frac{i}{\sigma_I}\mathrm{G}^\top \mathrm{b} = \mathrm{d}, \tag{12}$$

where d represents the vector of desired genetic gains, i corresponds to the selection intensity, σ_I is the standard deviation of the selection index, G represents the genetic covariances among traits, b is the vector of selection index weights, and $\mathrm{G}^\top$ denotes the transpose of matrix G. This methodology was later refined and formalized by Yamada et al.[28] and Itoh and Yamada[29]. The calculation of index coefficients follows the equation:

$$\mathrm{b} = \ \mathrm{P}^{-1}\mathrm{G}(\mathrm{GP}^{-1}\mathrm{G})^{-1}\mathrm{d}, \tag{13}$$

where P represents the phenotypic variance matrix among traits, and $(\mathrm{G\ P}^{-1}\mathrm{G})^{-1}$ is the generalized inverse ensuring that the coefficients are adjusted to optimize the correlated response among traits. We employed the `NLopt` package[30] using the `COBYLA` method (Constrained Optimization by Linear Approximations).

We selected the top 20% of genotypes to calculate the genetic gain ($\Delta G\%$) based on the Desired Gain Index, following Eq. (14).

$$\Delta G\% = \left(\frac{\hat{\mu_s} - \hat{\mu_o}}{\mu}\right) \times 100 \tag{14}$$

where $\hat{\mu_s}$ represents the mean BLUP of the selected genotypes, $\hat{\mu_0}$ is the mean BLUP of the evaluated population, and μ is the phenotypic mean. We used the ASReml-R[31] package for the analysis.

Results

The trials were conducted in 19 distinct environments in Kenya. Among them, Bungoma had the highest concentration, covering 31.6% of the total, followed by Murang'a (21.1%) and Njoro (15.8%). The remaining 31.5% was distributed across the other six locations (Table 1). Approximately two-third (68.4%) of the trials were sown in the first growing season, which typically begins between April and August, while 31.6% corresponded to the second growing season, with a sowing window between September and January. Overall, the mean values for GY, NDM, and PH were similar between the two seasons, with respective averages of 1899 and 1794 kg ha^{-1}, 125 and 111 days, and 67.4 and 47.8 cm. Three trials were conducted at altitudes above 2000 m, exclusively in Njoro, while three others were established below 1200 m, in the locations of Kibwezi (1178 m), Kirinyaga (1159 m), and Busia (1010 m), representing an altitudinal variation of approximately 1150 m.

Based on the boxplot, the traits PH, NDM, and GY exhibited median values of 59, 111, and 1630, respectively (Fig. 2). For GY, the overall mean was 1783.66 kg ha^{-1}, with the highest yield observed in E17 (2954.25 kg ha^{-1}) and the lowest in E12 (675.72 kg ha^{-1}), demonstrating high variability among environments (Table 1). For NDM, the scenario is similar. At E15, the genotypes reached maturity early, around 90 days, while at E05 in Njoro, this stage was achieved 98 days later. Despite this fluctuation, the overall mean across all environments was approximately 120 days (Table 1). Finally, a lower mean PH was observed at E08, Bungoma, with 34.61 cm. In contrast, taller plants were depicted in trials at Njoro (E02 and E05), the region with the highest altitude in the experiment network, with 107.18 cm and 105.15 cm respectively. However, in this study, the overall mean recorded for soybean monogenic trait (PH) was 62.35 cm (Table 1).

The LRT revealed significant effects ($p < 0.001$) of genotype, environment, and G×E interaction for all the three traits evaluated (Table 2). The estimated variance components indicated differences in the magnitude of genetic variability among the traits (Table 3). Phenotypic variance was higher for grain yield compared to PH and NDM. Broad-sense heritability (H_a^2) revealed that most of the phenotypic variance is explained by genetic variation. The genotype-environment correlation (r_{ge}^2) was high for all traits, indicating changes in the ranking of varieties across environments. For GY, the G×E interaction was large, indicating meaningful changes in genotype performance ranking across environments, which reflects a strong crossover interaction

Location	Environment	Year	Season	Trait		
				PH (cm)	NDM (days)	GY (kg ha^{-1})
Bungoma	E01	2019/20	1	71.84	115	2634.09
Njoro	E02	2019/20	1	107.18	182	2200.53
Kiambu	E03	2019/20	1	62.23	135	1773.83
Kibwezi	E04	2019/20	1	71.10	111	2331.35
Njoro	E05	2020/21	1	105.15	189	1108.54
Kirinyaga	E06	2020/21	1	61.16	113	1135.24
Muranga	E07	2020/21	1	60.52	117	2622.62
Bungoma	E08	2020/21	2	34.61	119	1212.83
Muranga	E09	2020/21	2	45.96	107	2641.98
Nakuru	E10	2020/21	2	53.03	122	1092.41
Siaya	E11	2020/21	2	56.21	103	1186.63
Bungoma	E12	2021/22	1	54.44	95	675.72
Njoro	E13	2021/22	1	86.64	176	2004.27
Muranga	E14	2021/22	1	57.29	104	1809.31
Bungoma	E15	2022/23	1	41.44	90	2044.71
Bungoma	E16	2022/23	2	44.61	101	1673.33
Muranga	E17	2022/23	2	52.47	113	2954.25
Bungoma	E18	2023/24	1	50.83	105	2710.91
Busia	E19	2023/24	1	46.22	97	1630.31
Overall mean				**62.35**	**120**	**1783.66**

Table 1. Summary of the overall means of grain yield (GY), number of days to maturity (NDM), and plant height at maturity (PH) in soybean trials across different environments in Kenya, Africa.

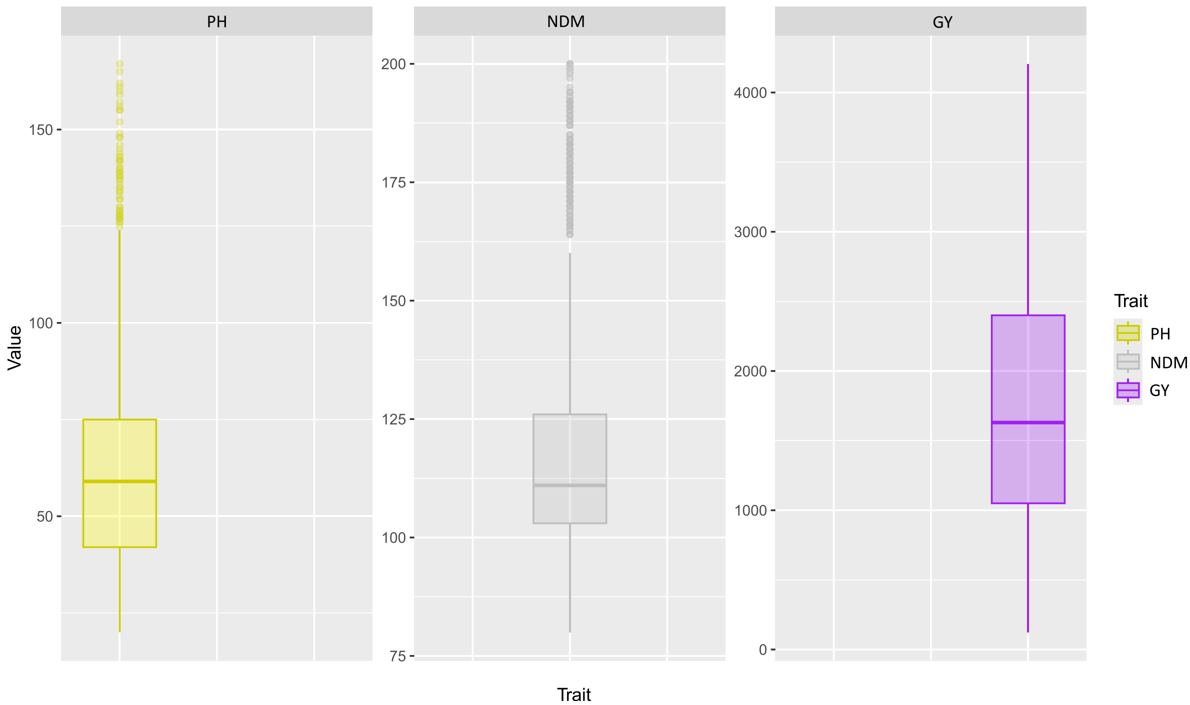

Fig. 2. Boxplot for plant height (PH, cm), number of days to maturity (NDM, days), and grain yield (GY, kg ha^{-1}) in soybean varieties evaluated in Kenya from the 2019 to 2023/24 crop seasons.

pattern. The CV_g indicated greater genetic variability for PH. Furthermore, high experimental precision was confirmed by the low values of CV_r. The CV_{ratio} indicated that genetic selection will be efficient for PH and NDM ($CV_{ratio} > 1$), whereas for GY, the residual variance exceeded the genotypic variance, making selection more challenging.

Source	LogL			LRT			χ^2		
	PH	NDM	GY	PH	NDM	GY	PH	NDM	GY
Gen	-6372.90	-4984.30	− 12359.00	333.44	235.38	42.94	$2e^{-16}$ ***	$2e^{-16}$ ***	$5.655e^{-11}$ ***
Env	-6438.10	− 4974.40	− 12382.00	130.49	215.70	89.71	$2e^{-16}$ ***	$2e^{-16}$ ***	$2.2e^{-16}$ ***
Gen × Env	-6552.60	− 5136.80	− 12573.00	359.47	540.55	471.77	$2e^{-16}$ ***	$2e^{-16}$ ***	$2.2e^{-16}$ ***

Table 2. Summary of the likelihood ratio test for plant height (PH, cm), number of days to maturity (NDM, days), and grain yield (GY, kg ha^{-1}) in 65 soybean varieties evaluated across 19 yield trials in Kenya from 2019 to 2023/24 crop seasons. LRT (Likelihood Ratio Test), LogL (Log-Likelihood), (χ^2) (Chi-Square Test), Gen (Genotypes), and Env (Environment)

Parameters	Traits		
	PH (cm)	NDM (days)	GY (kg ha^{-1})
$\hat{\sigma_g^2}$	4,281.30	70.42	75,785
$\hat{\sigma_p^2}$	778.1	991.00	928,856
r_{ge}^2	0.211	0.275	0.486
H_a^2	0.976	0.972	0.842
r_{ge}	0.518	0.657	0.584
CV_g	26.70	7.010	15.500
CV_r	15.40	3.480	22.200
CV_{ratio}	1.740	2.010	0.699

Table 3. Parameters for plant height (PH), number of days to maturity (NDM), and grain yield (GY, kg ha^{-1}) were estimated. These include genetic variance ($\hat{\sigma_g^2}$), phenotypic variance ($\hat{\sigma_p^2}$), broad-sense heritability (H_a^2), the proportion of variance due to G×E interaction (r_{ge}^2), G×E correlation (r_{ge}), genetic coefficient of variation (CV_g), residual coefficient of variation (CV_r), and the coefficient of variation ratio (CV_{ratio}).

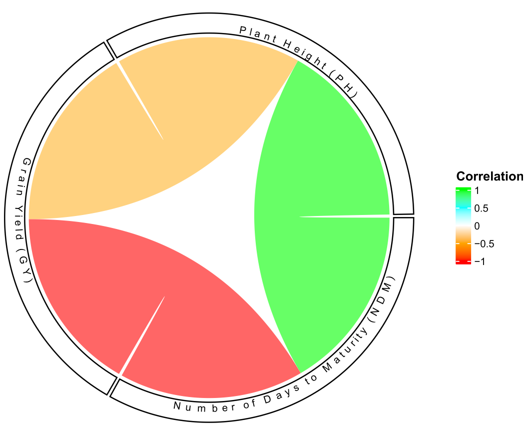

Fig. 3. Genetic correlations among soybean traits evaluated in Kenya during the cropping seasons from 2019 to 2023/24. The chord diagram illustrates the relationships between Plant Height (PH), Number of Days to Maturity (NDM), and Grain Yield (GY). The color gradient represents the strength and direction of the correlations: red indicates a strong negative correlation (−1), transitioning through orange (moderate negative), white (neutral, 0), cyan (moderate positive), and green for a strong positive correlation (+1). The width of the connections reflects the magnitude of the correlation. Only unique correlations are displayed to avoid redundancy.

Genetic correlation analysis indicated a moderate positive association between PH and NDM (r = 0.28), suggesting that taller plants tended to reach maturity later (Fig. 3). In contrast, GY showed a strong negative correlation with NDM (r = − 0.71), implying that late-maturing genotypes were generally less productive. Additionally, PH and GY exhibited a moderate negative correlation (r = – 0.42), indicating that increased plant height did not translate into higher yields (Fig. 3). These findings highlight the importance of balancing plant architecture and productivity in soybean breeding for Kenyan conditions.

Based on the BLUP values for the evaluated traits, specific genotype performance was observed (Fig. 4). For PH, G29, G11, G48, G07, and G01 exhibited the shorter heights. Regarding NDM, genotypes G10, G32, G09, G36, and G34 had a shorter cycle, maturing earlier than the others. For GY, the highest-yielding genotypes were G42, G26, G46, G35, and G53 respectively. In contrast, G21, G22, G23, and G24 were characterized by taller plant heights, longer growing cycles, and lower grain yield, making them less desirable for selection.

The Biplot analysis (Fig. 5) categorized genotypes and environments based on overall performance and stability. Quadrant I comprised of unstable genotypes in highly discriminative environments but with trait values below the overall mean. Quadrant II included unstable genotypes (high WAASB) with performance above the mean in environments with strong discriminative ability. In Quadrant III, genotypes exhibited high stability but low overall performance, while the environments had limited discriminative capacity. Quadrant IV encompassed high-performing and stable genotypes (low WAASB), making them the most suitable for traits targeted for improvement, such as GY. For PH and NDM, selection prioritized stable genotypes with lower trait values, highlighting G37, G58, G29, and G17 for PH and G10, G36, G32, and G13 for NDM. Regarding GY, the focus was on identifying genotypes within Quadrant IV to maximize productivity, with G42 as the highest-yielding genotype, exceeding the overall mean.

Concerning the environments, although E17 (Muranga, 1355 m, 2022/23) stood out for having the highest average yield among all experiments (Table 1), its position in the graph (Fig. 5C) indicates a low ability to discriminate genotypes. On the other hand, E12 (Bungoma, 1517 m, 2021/22), which recorded the lowest average yield, was positioned in the first quadrant of the WAASB graphs for PH (Fig. 5A) and GY (Fig. 5C), suggesting that this region has a high discriminative ability for genotypes (except for NDM), a pattern similar to that observed in E05 (Njoro, 2162 m, 2020/21). A comparable performance was observed in E02 (Njoro, 2162 m, 2019/20), where the materials exhibited greater height compared to the others (Table 1). This environment also demonstrated a high discriminative ability due to its location in quadrant II. However, it may be less desirable, as the PH and NDM values in this environment were above the average (Fig. 5).

Twelve (12) genotypes; G42, G26, G46, G35, G53, G41, G12, G54, G39, G52, G02, and G37, were selected based on the Desired Gain Index. Among them, two genotypes, G42 and G26, exhibited the highest index values, being considered the most desirable ideotypes for soybean under Kenya's diverse soybean growing conditions (Fig. 6).

The density distribution of selected and non-selected genotypes for the variables PH, NDM, and GY is presented in Fig. 7. It is observed that the selected genotypes (in green) tend to shift compared to the non-selected ones (in red), reflecting the impact of selection based on the Desired Gain Index (Fig. 7). Specifically, the selected genotypes exhibited a genetic gain of 6.5% for plant height (PH, Fig. 7A), 1.3% for days to maturity (NDM, Fig. 7 B), and 15.6% for grain yield (GY, Fig. 7C). These results indicate that applying the index enabled the identification of superior genotypes based on the established selection criteria.

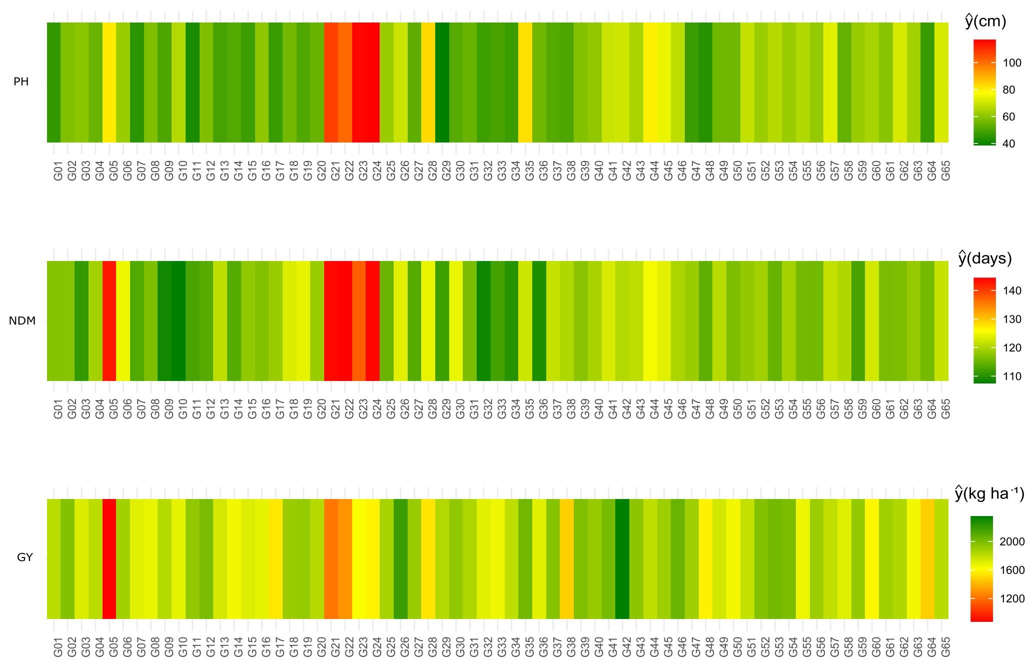

Fig. 4. Visualization of BLUP values for soybean genotypes related to plant height (PH), days to maturity (NDM), and grain yield (GY). Darker shades of green indicate genotypes with more favorable BLUP values, while stronger shades of red represent genotypes with less desirable trait performance.

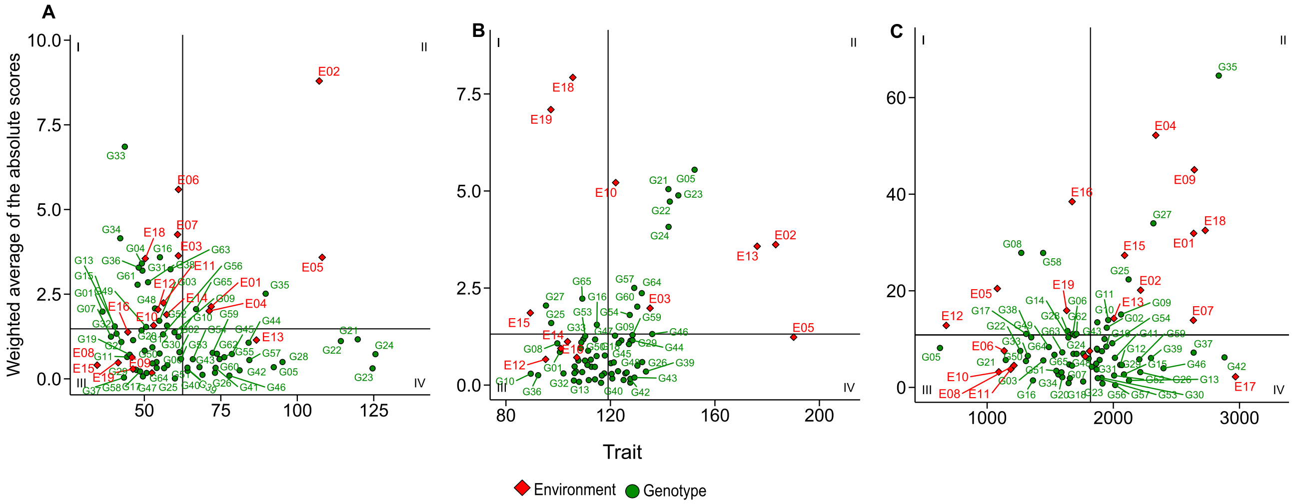

Fig. 5. The figure presents scatter plots illustrating the relationship between different phenotypic variables and the weighted average of absolute scores in a plant breeding experiment. In panel 1(**A**) the relationship between plant height (PH) and the weighted average of absolute scores is shown, while panel 1(**B**) presents the relationship between days to maturity (NDM) and the weighted average of absolute scores. In panel 1(**C**) grain yield (GY) is correlated with the weighted average of absolute scores. Green points represent the evaluated genotypes, whereas red points correspond to different experimental environments. The vertical and horizontal lines indicate the mean limits of the variables, allowing the identification of genotypes and environments with contrasting performances.

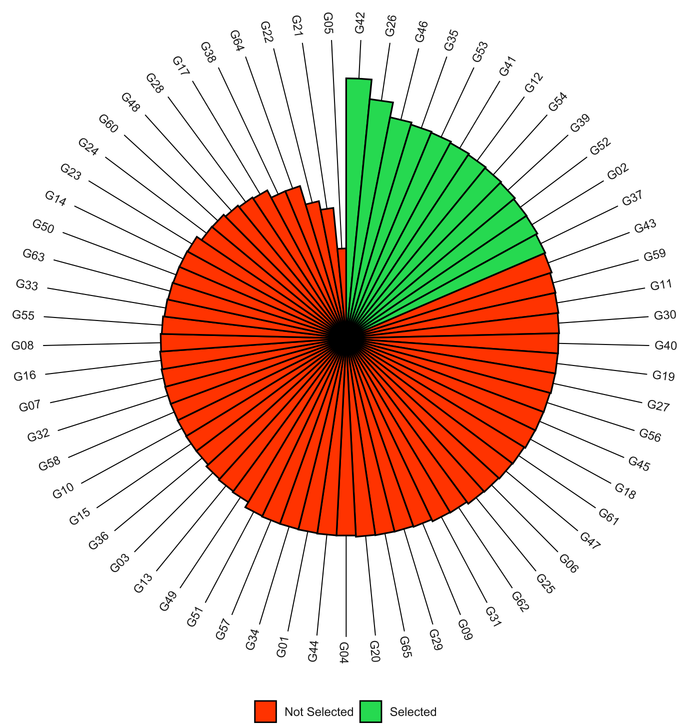

Fig. 6. The circular bar chart illustrates the selected (green) and non selected (red) genotypes. The bar length represents the genotype's superiority across the three traits based on the desired genetic index.

Discussion

This study evaluated 65 soybean genotypes across 19 environments in Kenya and revealed substantial variability for PH, NDM, and GY. The LRT demonstrated significant effects of genotype, environment, and G×E interaction for all traits. High-performing and stable genotypes were identified using a multi-trait selection index, leading to genetic gains of 6.5% for PH, 1.3% for NDM, and 15.6% for GY. Twelve genotypes stood out for

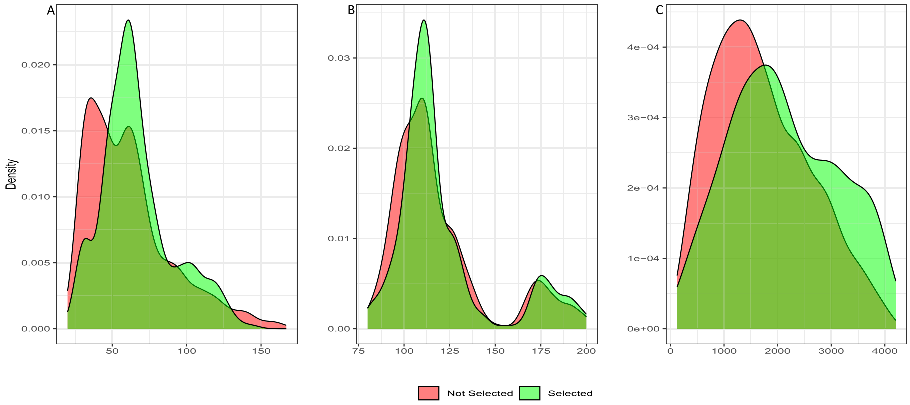

Fig. 7. Density plot illustrating the distribution of phenotypic traits: (**A**) plant height (PH), (**B**) number of days to maturity (NDM), and (**C**) grain yield (GY). The red normal curve represents the mean of the original population (μ_0), while the green curve indicates the mean of the selected population (μ_s).

their superior performance and adaptability under Kenyan growing conditions. As part of the Pan-African Trials (PATs), our study applied multi-environment trials (METs) across a broad range of environments, enabling the simultaneous assessment of multiple traits and genotypes under contrasting conditions of climate, soil, altitude, and management. Such trials are essential for evaluating the reliability and performance of candidate varieties, as emphasized by Gesteira et al. [32]. The key objective in this context is to expand METs across different target populations of environments (TPEs), seasons, and years, while broadening the diversity of assessed genotypes and facilitating the identification and recommendation of those best suited to new agroecological zones[33–35]. However, a significant challenge in this process is the strong influence of crossover G×E interactions, which alter the ranking of varieties across sites and complicate the selection of candidates that combine high performance and stability in agronomic traits, as highlighted in Table 2.

Throughout the selection process, genotypes with poor adaptation to the TPEs, lower GY, or undesirable agronomic characteristics are systematically discarded, while new experimental genotypes are continually introduced into the evaluation pipeline. This dynamic reprocessing commonly leads to genetic and statistical imbalance in multi-environment trials, as highlighted by Araújo et al. [36]. Such imbalance becomes more critical in networks that involve highly contrasting environments, as is the case in Kenya. The strong environmental heterogeneity typically intensifies G×E interaction, which poses a major challenge for breeders when selecting genotypes that combine high performance with stability. In particular, previous studies have demonstrated that key environmental factors–such as altitude, maximum temperature, and soil fertility–exert significant and distinct impacts on G×E interaction patterns[37]. In this context, altitude has been consistently reported as a major driver of variation in soybean development, affecting phenological traits such as time to maturity and plant height. This is largely attributed to genotype × altitude interactions, which tend to have a stronger effect on phenotypic expression than genotype × latitude interactions[38]. Such environmental influences not only modify the absolute performance of genotypes but also contribute to crossover interactions, leading to changes in genotype ranking across environments. Similar patterns of significant G×E interaction for soybean traits, especially grain yield, have been widely documented in studies from various countries and continents, including Africa, Asia, and the Americas[39–43]. In Africa, research has been more concentrated in Southern Africa[33,44], as well as in other regions such as Ethiopia[45,46] and Nigeria[47]. These findings reinforce the importance of using METs combined with robust statistical models to properly account for environmental variation, improve genotype evaluation, and support more accurate selection decisions in breeding programs targeting diverse and challenging production environments.

The results indicate that the phenotypic variance was substantially higher for GY compared to PH and NDM (Table 3), highlighting the complex nature of this trait and its strong dependence on environmental conditions[48,49]. Moreover, the r^2_{ge} values were moderate to high across all traits (Table 3), confirming a strong G×E interaction and considerable changes in genotype rankings across environments. Furthermore, the genotypic correlation among environments r^2_{ge} ranged from 0.518 to 0.657 (Table 3) for all traits investigated. A high r_{ge} facilitates the identification of stable and superior genotypes, suggesting a consistent pattern across environments. Conversely, a low r_{ge} presents a challenge for selecting such genotypes, as it indicates greater variability in their performance across different conditions[50]. Therefore, integrating strategies that explicitly account for G×E–such as METs–and incorporating multi-trait selection is essential for identifying genotypes not only with superior agronomic performance but also with broad adaptability and stability across diverse environments[51]. The high estimates of H^2_a for all traits suggest that a large portion of the phenotypic variance is attributable to genetic factors, reinforcing the potential for genetic gain through selection and the feasibility of maintaining progress via recurrent selection[52]. In this regard, METs provide an optimal framework for assessing heritability–a key parameter reflecting the reliability of observed phenotypes–across varying environmental conditions, particularly through the application of eBLUP, which enhances the precision of its estimates and enables the accurate selection of genetically superior individuals[53–55].

One of the major challenges in plant breeding is managing unfavorable correlations between traits, which can hinder progress simultaneous genetic improvement. This complexity was manifest in our study through the analysis of genetic correlations among the evaluated traits, where plant height (PH) showed a positive correlation with NDM ($r = 0.28$), suggesting that taller genotypes tend to exhibit longer growth cycles, which may be an undesirable combination in production systems that require early-maturing cultivars with low lodging risk. More notably, GY was negatively correlated with both PH ($r = -0.42$) and NDM ($r = -0.71$), signaling that taller or late-maturing genotypes generally yielded less, possibly due to prolonged exposure to adverse conditions during the reproductive phase[56], similar findings were reported in soybean by Bhartiya et al.[57], Nataraj et al.[58], and Wang et al.[59]. This behavior was markedly evident in genotype G05, which exhibited the lowest overall yield (669.47 kg ha^{-1}) while presenting the longest growth cycle (148 days to R8). These findings align with previous studies conducted in China, where architectural traits such as plant height, number of nodes, and branches showed significant correlation with maturity stages, supporting that these genetic relationships, often driven by pleiotropy or linkage, impose constraints on the simultaneous optimization of desirable traits[60].

Further supporting these results, the breeder's equation, expressing genetic gain as the product of selection accuracy, genetic variation, and selection intensity over time[61], highlights how such unfavorable correlations can diminish long-term breeding efficiency[53,61]. In your database, PH showed the highest CV_g, indicating a greater genetic diversity and higher proportional gain potential, while NDM had the lowest CV_r, reflecting a higher experimental precision in its measurement. The CV_{ratio} values greater than 1.0 for PH and NDM points to a predominance of genetic over residual variance, whereas a value below 1.0 for GY suggests stronger environmental influence, potentially limiting the accuracy of selection for this trait. To neutralize these negative correlations, breeders may imposed other selection methods such as independent culling, tandem selection and index selection, enabling the simultaneous improvement of multiples features.

In this context, given the high G×E interaction detected in our work (Tables 2 and 3), the use of traditional frequentist models, such as ANOVA based on the ordinary least squares method, may present restrictions, particularly under the genetic imbalance and environmental heterogeneity of METs. To address these bottlenecks and improve the accuracy of genotype assessment, we employed the WAASB model, which uses a linear mixed-effects model that effectively captures the G×E patterns. Unlike other stability statistics such as AMMI Stability Value (ASV), its main advantages include the robust estimates using the weighted average of absolute deviations of the interaction principal components (IPCA) scores,instead of squared deviations, making it less sensitive to outliers[26,62]. This was crucial in our dataset, where significant crossover interactions were present and genotype varied across environments. The WAASB method enabled the identification of varieties such as G42 that merge high performance and broad adaptability (Fig. 5), confirming its suitability for stability analysis in the context of METs[63–65].

The integration of biplots combining WAASB and GY (or another trait) facilitates the visualization of genotypes that achieve a balance between high mean performance and stability (Supplementary Figure 2), as observed in the present study with genotype G42 (Fig. 5). In addition to GY, other agronomic traits such as PH and NDM may also contribute to genotype selection. In this regard, genotypes showing relatively lower PH and shorter NDM–such as G37, G58, G29, and G17 (for PH), and G10, G36, G32, and G13 (for NDM)–could be of interest, as these traits are often associated with improved agronomic performance, better harvestability, and potential adaptability to environments with shorter growing seasons or stress-prone conditions[66]. With respect to the environments, E15 (Bungoma, 1526 m, 2022/2023) stands out not only for its greater earliness compared to the others but also for its strong discriminative ability for GY and NDM, like E12 for PH (Fig. 5). This facilitates genotype selection in these regions, showcasing their necessity in soybean METs[45,58]. Thus, WAASBY proved to be an effective tool for selecting genotypes with desirable trait values and broad adaptability, as reported in previous studies on soybean[33,47,58,67,68].

To address this complexity and enhance selection efficiency, breeders commonly adopt selection indices, which consolidate multiple traits into a single value per genotype. This strategy enables more efficient and integrated selection decisions[16,69]. The choice of selection index depends on specific breeding objectives, trait heritability, population structure, and developmental stage of the program. In this context, selection indices have proven to be effective in various crops such as soybean[32,70,71], common bean[72], and wheat[73], with their suitability shaped by genetic correlations and breeding strategies[74]. Among the various methods, we adopted the Desired Gain Index (DGI) proposed by Pesek and Baker[15], which is particularly valuable due to its flexibility in specifying expected genetic gains for each trait. Empirical studies confirm the superiority of index-based approaches over direct or single-trait selection, especially when dealing for complex traits like grain yield[70,73]. In our study, the selection intensity of 20% applied to the calculation of the Desire Index enabled the identification of 12 desirable genotypes, from which G42 and G26 standing out in decreasing order of relevance. Unlike the Smith-Hazel index, which focuses on profit maximization, our approach aimed to enhance the expected genetic gains[75] by selecting varieties with high performance and stability across the 19 environments evaluated in Kenya, Africa.

As a result, we observed genetic gains of 1.3% for NDM, 6.5% for PH, and 15.6% for GY, ensuring continuous progress in population improvement over time[75] and allowing for the recommendation of specific genotypes for different specific regions. Taken together, we demonstrated that the WAASB model efficiently selects soybean genotypes with high stability and broad adaptation[26] to Kenyan agro-climatic conditions. Moreover, we showed that the Desired Gain Index is a robust and effective tool for identifying superior ideotypes, particularly in multi-trait selection scenarios. This approach reduces the inherent subjectivity associated with assigning economic weights to each trait[27–29]. Our primary goal was to simultaneously evaluate grain yield, plant height, and physiological maturity, to facilitate an objective and accurate selection of ideotypes best suited to the diverse local agroecological conditions in Kenya. This strategy has the potential to directly benefit small and medium-sized Kenyan farmers by enabling them to incorporate genetically superior and locally adapted materials into their production systems. Thus, prior knowledge of genotype performance and stability empowers farmers to make strategic and informed decisions in improving their agricultural sustainability and profitability while mitigating the risks associated with climate change, which is increasingly becoming a global agricultural crisis.

Conclusion
This study demonstrated a strong genotype-by-environment interaction for soybean under Kenya's diverse edaphoclimatic conditions. Multi-trait selection enabled the identification of superior genotypes, with genetic gains of 6.5% in plant height, 1.3% in days to maturity, and 15.6% in grain yield. Genotypes G42 and G26 stood out as the most desirable ideotypes for Kenya, offering promising options for advancing soybean breeding in the region.

Data availability
The phenotypic data analyzed during the current study are available from the supplementary material.

Received: 17 April 2025; Accepted: 4 July 2025
Published online: 29 July 2025

References
1. Pereyra, V. M., Hefley, T., Prasad, P. V. & Ciampitti, I. A. Soybean seed yield, protein, and oil concentration for a modern and old genotype under varying row spacings. *Heliyon* **10**, e35054. https://doi.org/10.1016/j.heliyon.2024.e35054 (2024).
2. Nget, R. et al. Responses of soybean genotypes to different nitrogen and phosphorus sources: Impacts on yield components, seed yield, and seed protein. *Plants* **11**, 298. https://doi.org/10.3390/plants11030298 (2022).
3. Bellaloui, N. et al. Agricultural practices altered soybean seed protein, oil, fatty acids, sugars, and minerals in the Midsouth USA. *Front. Plant Sci.* **6**, 31. https://doi.org/10.3389/fpls.2015.00031 (2015).
4. Diers, B. & Scaboo, A. Soybean breeding in Africa. *Afr. J. Food Agric. Nutr. Dev.***19**, 15121–15125. https://doi.org/10.18697/ajfand.88.SILFarmDoc03 (2019).
5. Santos, M. Soybean varieties in sub-saharan Africa. *Afr. J. Food Agric. Nutr. Dev.***19**, 15136–5139. https://doi.org/10.18697/ajfand.88.SILFarmDoc06 (2019).
6. Malosetti, M., Ribaut, J.-M. & van Eeuwijk, F. A. The statistical analysis of multi-environment data: Modeling genotype-by-environment interaction and its genetic basis. *Front. Physiol.* **4**, 44. https://doi.org/10.3389/fphys.2013.00044 (2013).
7. Isik, F. et al. Multi environmental trials. *Genetic Data Anal. Plant Animal Breed.* 227–262 (2017).
8. Araújo, M. S. et al. Evaluation of adaptability and stability for iron, zinc and protein content in cowpea genotypes using GGE biplot approach. *Heliyon* **8**, e11832. https://doi.org/10.1016/j.heliyon.2022.e11832 (2022).
9. Poupon, V., Gezan, S. A., Schueler, S. & Lstiburek, M. Genotype × environment interaction and climate sensitivity in growth and wood density of European larch. *For. Ecol. Manag.* **545**, 121259. https://doi.org/10.1016/j.foreco.2023.121259 (2023).
10. Lynch, M. & Walsh, B. *Genetics and Analysis of Quantitative Traits* (Sinauer Associates Inc, 1998).
11. Alves, R. S. et al. Multiple-trait BLUP in longitudinal data analysis on *Jatropha curcas*. *Indus. Crops Manag.* **545**, 121259. https://doi.org/10.1016/j.indcrop.2018.12.019 (2023).
12. Melo, V. et al. Modeling (co)variance structures for genetic and non-genetic effects in the selection of common bean progenies. *Euphytica* **216**, 1–13. https://doi.org/10.1007/s10681-020-02607-9 (2020).
13. Araújo, M. S. et al. Multi-trait selection for nutritional and physiological quality of cacao genotypes in irrigated and non-irrigated environments. *Sci. Rep.* **14**, 6368. https://doi.org/10.1038/s41598-024-56556-7 (2024).
14. Hazel, L. The genetic basis for constructing selection indexes. *Genetics* **28**, 476–490 (1943).
15. Pesek, J. & Baker, R. Desired improvement in relation to selection indices. *Can. J. Plant Sci.* **49**, 803–804. https://doi.org/10.4141/cjps69-137 (1969).
16. Olivoto, T. & Nardino, M. MGIDI: Toward an effective multivariate selection in biological experiments. *Bioinformatics* **37**, 1383–1389. https://doi.org/10.1093/bioinformatics/btaa981 (2021).
17. Araújo, M. S. et al. Optimizing soybean variety selection for the pan-African trial network using factor analytic models and envirotyping. *Front. Plant Sci.* **16**, 1594736. https://doi.org/10.3389/fpls.2025.1594736 (2025).
18. R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing (version 4.5.1), Vienna, Austria (2025).
19. Pebesma, E. Simple features for r: Standardized support for spatial vector data. *The R Journal* **10**, 439–446 (2018).
20. Hijmans, R. J. *terra: Spatial Data Analysis* (2024). R package version 1.8-42.
21. Wickham, H. *ggplot2: Elegant Graphics for Data Analysis* (Springer-Verlag, New York, 2016).
22. South, A. *rnaturalearth: World Map Data from Natural Earth* (2017). R package version 0.1.1.
23. Patterson, H. D. & Thompson, R. Recovery of inter-block information when block sizes are unequal. *Biometrika* **58**, 545–554. https://doi.org/10.1093/biomet/58.3.545 (1971).
24. Henderson, C. R. Estimates of changes in herd environment. *J. Dairy Sci.* **61**, 294–300 (1949).
25. Wilks, S. S. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *Ann. Math. Stat.* **9**, 60–62 (1938).
26. Olivoto, T. et al. Mean performance and stability in multi-environment trials I: Combining features of AMMI and BLUP techniques. *Agron. J.* **111**, 2949–2960. https://doi.org/10.2134/agronj2019.03.0220 (2019).
27. Pešek, J. & Baker, R. J. Desired improvement in relation to selection indices. *Can. J. Plant Sci.* **49**, 803–804. https://doi.org/10.4141/cjps69-137 (1969).
28. Yamada, Y., Yokouchi, K. & Nishida, A. Selection index when genetic gains of individual traits are of primary concern. *Jpn. J. Genet.* **50**, 33–41. https://doi.org/10.1266/jjg.50.33 (1975).
29. Itoh, Y. & Yamanda, Y. Re-examination of selection index for desired gains. *Genet. Sel. Evol.* **18**, 499. https://doi.org/10.1186/1297-9686-18-4-499 (1986).
30. Johnson, S. G. *The NLopt Nonlinear-Optimization Package* (2008). Accessed: 2025-03-07.
31. Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. & Thompson, R. *ASReml-R Reference Manual Version 4.2*. VSN International Ltd., Hemel Hempstead, HP2 4TP, United Kingdom (2023).
32. Gesteira, G. S., Bruzi, A. T., Zito, R. K., Fronza, V. & Arantes, N. E. Selection of early soybean inbred lines using multiple indices. *Crop Sci.***58**, 2494–2502. https://doi.org/10.2135/cropsci2018.05.0295 (2018).
33. Meyer, E. et al. Evaluating genetic diversity and seed composition stability within pan-African soybean variety trials. *Crop Sci.* **64**, 3272–3292. https://doi.org/10.1002/csc2.21356 (2024).
34. Werner, C. R., Zaman-Allah, M., Assefa, T., Cairns, J. E. & Atlin, G. N. Accelerating genetic gain through early-stage on-farm sparse testing. *Trends Plant Sci.* **30**, 17–20. https://doi.org/10.1016/j.tplants.2024.10.010 (2024).
35. Crespo-Herrera, L. A. et al. Target population of environments for wheat breeding in India: Definition, prediction and genetic gains. *Front. Plant Sci.* **12**, 638520. https://doi.org/10.3389/fpls.2021.638520 (2021).

36. Araújo, M. S. et al. Modeling covariance structures for genetic and non-genetic effects in cowpea multi-environment trials. *Agron. J.* **115**, 1248–1256. https://doi.org/10.1002/agj2.21321 (2023).
37. Batista, A. B. d. O. J., Chaves, L. J., Couto, M. A. et al. Environmental and genotypic factors associated with genotype by environment interactions in soybean. *Crop Breed. Appl. Biotechnol.* **6**, 79-86 (2006).
38. Alliprandini, L. F. et al. Understanding soybean maturity groups in Brazil: Environment, cultivar classification, and stability. *Crop Sci.* **49**, 801–808. https://doi.org/10.2135/cropsci2008.07.0390 (2009).
39. Silva, W. J. D. S. et al. Yield of soybean genotypes identified through GGE biplot and path analysis. *Plos One* **17**, e0274726. https://doi.org/10.1371/journal.pone.0274726 (2022).
40. Wijaya, A. A. et al. Grain yield stability of black soybean lines across three agroecosystems in West Java, indonesia. *Open Agric.* **7**, 749–763. https://doi.org/10.1515/opag-2022-0137 (2022).
41. Yan, W. & Rajcan, I. Biplot analysis of test sites and trait relations of soybean in Ontario. *Crop Sci.* **42**, 11–20. https://doi.org/10.2135/cropsci2002.1100 (2002).
42. Agoyi, E. E. et al. Genotype by environment effects on promiscuous nodulation in soybean (*Glycine max* (L.) Merrill). *Agric. Food Sec.* **6**, 1–14. https://doi.org/10.1186/s40066-017-0107-7 (2017).
43. Nataraj, V. et al. Envirotype-based delineation of environmental effects and genotype × environment interactions in Indian soybean (*Glycine max* (L.). *Sci. Rep.***14**, 11629, https://doi.org/10.1038/s41598-024-62613-y (2024).
44. Mwiinga, B., Sibiya, J., Kondwakwenda, A., Musvosvi, C. & Chigeza, G. Genotype x environment interaction analysis of soybean (*Glycine max* (L.) Merrill) grain yield across production environments in Southern Africa. *Field Crops Res.* **256**, 107922. https://doi.org/10.1016/j.fcr.2020.107922 (2020).
45. Hailemariam Habtegebriel, M. Adaptability and stability for soybean yield by AMMI and GGE models in Ethiopia. *Front. Plant Sci.* **13**, 950992. https://doi.org/10.3389/fpls.2022.950992 (2022).
46. Habtegebriel, M. H. & Abebe, A. T. Grain yield stability of soybean (*Glycine max* (L.) Merrill) for different stability models across diverse environments of Ethiopia. *Agrosyst. Geosci. Environ.* **6**, e20396. https://doi.org/10.1002/agg2.20396 (2023).
47. Abebe, A. T. et al. Genotype × environment interaction and yield stability of soybean (*Glycine max* L.) genotypes in multi-environment trials (METs) in Nigeria. *Heliyon* **10**, e38097. https://doi.org/10.1016/j.heliyon.2024.e38097 (2024).
48. Ray, D. K., Gerber, J. S., MacDonald, G. K. & West, P. C. Climate variation explains a third of global crop yield variability. *Nat. Commun.* **6**, 5989. https://doi.org/10.1038/ncomms6989 (2015).
49. Iizumi, T. & Ramankutty, N. Changes in yield variability of major crops for 1981–2010 explained by climate change. *Environ. Res. Lett.* **11**, 034003. https://doi.org/10.1088/1748-9326/11/3/034003 (2016).
50. Koundinya, A. et al. Genetic parameters, stability and selection of cassava genotypes between rainy and water stress conditions using AMMI, WAAS BLUP and MTSI. *Sci. Horticult.* **281**, 109949. https://doi.org/10.1016/j.scienta.2021.109949 (2021).
51. Sampaio Filho, J. S., Olivoto, T., Souza Campos, M. & de Oliveira, E. J. Multi-trait selection in multi-environments for performance and stability in cassava genotypes. *Front. Plant Sci* **14**, 1282221, https://doi.org/10.3389/fpls.2023.1282221 (2023).
52. Silva, É. M. d. et al. Genetic parameters and selection for multiple traits in recurrent selection populations of maize. *Rev. Ceres* **70**, 81–90. https://doi.org/10.1590/0034-737X202370030009 (2023).
53. Pour-Aboughadareh, A., Khalili, M., Poczai, P. & Olivoto, T. Stability indices to deciphering the genotype-by-environment interaction (GEI) effect: An applicable review for use in plant breeding programs. *Plants* **11**, 414. https://doi.org/10.3390/plants11030414 (2022).
54. Ramalho, M. A. P., Carvalho, B. L. & Nunes, J. A. R. Perspectives for the use of quantitative genetics in breeding of autogamous plants. *Int. Schola. Res. Notices* **2013**, 718127. https://doi.org/10.5402/2013/718127 (2013).
55. Smith, A., Cullis, B. R. & Thompson, R. The analysis of crop cultivar breeding and evaluation trials: An overview of current mixed model approaches. *J. Agric. Sci.* **143**, 449–462. https://doi.org/10.1017/S0021859605005587 (2005).
56. Messina, C., Hammer, G., Dong, Z., Podlich, D. & Cooper, M. Modelling crop improvement in a G×E×M framework via gene-trait-phenotype relationships. in *Crop Physiology: Applications for Genetic Improvement and Agronomy*, 235–265. https://doi.org/10.1016/B978-0-12-374431-9.00010-4 (2009).
57. Bhartiya, A. & Aditya, J. Genetic variability, character association and path analysis for yield and component traits in black seeded soybean lines under rainfed condition of Uttarakhand hills of India. *Legume Res. Int. J.* **39**, 31–34. https://doi.org/10.0.73.117/lr.v0iOF.7482 (2016).
58. Nataraj, V. et al. WAASB-based stability analysis and simultaneous selection for grain yield and early maturity in soybean. *Agron. J.* **113**, 3089–3099. https://doi.org/10.1002/agj2.20750 (2021).
59. Wang, B. et al. Genetic variation analysis, correlation analysis and principal component analysis on agronomic traits of summer sowing soybean (*Glycine max* Merr.) in Huang-Huai-hai region. *Agric. Biotechnol.* **2**, 2164-4993 (2013).
60. Zhu, X. et al. The correlation between maturity, plant architecture and yield related traits of soybean cultivars in Northern China. *Legume Res.* **47**, 2095–2102. https://doi.org/10.18805/LRF-822 (2024).
61. Cobb, J. N. et al. Enhancing the rate of genetic gain in public-sector plant breeding programs: Lessons from the breeder's equation. *Theor. Appl. Genet.* **132**, 627–645. https://doi.org/10.1007/s00122-019-03317-0 (2019).
62. Olivoto, T. et al. Mean performance and stability in multi-environment trials. II. *Agron. J.* **111**, 2961–2969. https://doi.org/10.2134/agronj2019.03.0221 (2019).
63. Resende, M. D. V. & Alves, R. S. Linear, generalized, hierarchical, bayesian and random regression mixed models in genetics/genomics in plant breeding. *Funct. Plant Breed. J.* **2**, 121-152. https://doi.org/10.35418/2526-4117/v2n2a1 (2020).
64. Rosa, J. C. et al. A new Bayesian approach to the Toler model for evaluating the adaptability and stability of genotypes. *Crop Breed. Appl. Biotechnol.* **23**, e440523111. https://doi.org/10.1590/1984-70332023v23n1a11 (2023).
65. Nuvunga, J. J. et al. Stability and adaptability of elite upland rice lines using Bayesian-AMMI model. Austr. J. Crop Sci. 15, 244–250, https://doi.org/10.21475/ajcs.21.15.02.p2882 (2021)
66. Miao, L. et al. What factors control plant height?. *J. Integr. Agric.* **23**, 1803–1824. https://doi.org/10.1016/j.jia.2024.03.058 (2024).
67. Abdelghany, A. M. et al. Exploring the phenotypic stability of soybean seed compositions using multi-trait stability index approach. *Agronomy* **11**, 2200, https://doi.org/10.3390/agronomy11112200 (2021).
68. Rajput, L. et al. WAASB index revealed stable resistance sources for soybean anthracnose in India. *J. Agric. Sci.* **159**, 710–720. https://doi.org/10.1017/S0021859622000016 (2021).
69. Jahufer, M. & Casler, M. Application of the smith-hazel selection index for improving biomass yield and quality of switchgrass. *Crop Sci.* **55**, 1212–1222. https://doi.org/10.2135/cropsci2014.08.0575 (2015).
70. Bizari, E. H. et al. Selection indices for agronomic traits in segregating populations of soybean. *Rev. Cienc. Agron.* **48**, 110–117. https://doi.org/10.5935/1806-6690.20170012 (2017).
71. Soares, I. O. et al. Adaptability of soybean cultivars in different crop years. *Genet. Mol. Res.* **14**, 8995–9003. https://doi.org/10.4238/2015.August.7.8 (2015).
72. Jost, E. et al. Comparison among direct, indirect and index selections on agronomic traits and nutritional quality traits in common bean. *J. Sci. Food Agric.* **93**, 1097–1104. https://doi.org/10.1002/jsfa.5856 (2013).
73. Fellahi, Z. E. A., Hannachi, A. & Bouzerzour, H. Analysis of direct and indirect selection and indices in bread wheat (*Triticum aestivum* L.) segregating progeny. *Int. J. Agron.***2018**, 8312857, https://doi.org/10.1155/2018/8312857 (2018).
74. Céron-Rojas, J. J. & Crossa, J. *Linear Selection Indices in Modern Plant Breeding* (Springer Nature, 2018).
75. Werner, C. R., Gardner, K. A. & Tolhurst, D. J. Reviving the desired gains index: An optimal solution for parent selection in public plant breeding programs. *bioRxiv* 2024–07, https://doi.org/10.1101/2024.07.21.603926 (2024).

Acknowledgements
This research was supported by the Coordination for the Improvement of Higher Education Personnel (CAPES) and São Paulo Research Foundation (FAPESP, Grant 2024/01868), National Council for Scientific and Technological Development (CNPq), Soybean Innovation Lab (SIL), and Luiz de Queiroz College of Agriculture of the University of São Paulo.

Author contributions
Conceived the experiment(s): [A.A.A., J.P.S.P., E.P.L.]; Conducted the experiment(s): [P.G., G.C., M.F.S., T.G., J.N]. Provided insights into the methodology: [M.S.A., B.F.F., B.W.D.]; Analyzed the results: [A.A.A., J.P.S.P., M.S.A.]; Writing—review and editing: [A.A.A., J.P.S.P., M.S.A., I.V.U.]; Provided critical revisions of the paper drafts [I.V.U., J.B.P.]. All authors reviewed and approved the final version of the manuscript.

Declarations

Competing interests
The authors declare that they have no conflicts of interest.

Ethical approval
The plant material comprises soybean varieties cultivated in Africa by the Soybean Innovation Lab (SIL). We complied with all necessary regulations and guidelines required to conduct this research.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-10654-2.

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